

Robustness Study

Noise, synthetic defects, and out-of-distribution probes

PAUT Explainable-AI Digital Twin — weld defect inspection from Total Focusing Method (TFM) images.

Generated by `scripts/robustness.py`. A separate stress test — it does NOT change the reported train/test accuracy. Three probes: additive noise, synthetic defects, and a text out-of-distribution false-positive check.

A. Noise robustness

Gaussian noise added to real test images at rising σ ; segmentation Dice and type accuracy measured at each level.

σ	Foreground Dice	Type accuracy
0.00	0.594	1.000
0.05	0.509	0.500
0.10	0.580	0.500
0.20	0.052	0.500
0.40	0.007	0.500

Curve: data/processed/robustness/noise_curve.png.

B. Synthetic defects

Procedurally generated porosity (round blobs) and slag (streak) images (16 total). The model **detected a defect in 100%** of them; type accuracy 50%. Gallery: data/processed/robustness/synthetic_gallery.png.

C. Text probe (false-positive / OOD)

Plain dark images with white text (no real defect). The model flagged a (false) defect in **0%** of the 6 text images (robust — mostly ignores text). Gallery: data/processed/robustness/text_gallery.png.

Findings & interpretation

- **Noise is the weak point.** Foreground Dice falls from 0.59 (clean) to 0.01 at $\sigma=0.40$, and type accuracy drops to chance (0.50) by $\sigma=0.05$ — the scattering type-classifier was trained on clean features, so additive noise pushes them out of distribution and it collapses to one class. **Mitigation:** add additive-noise augmentation when extracting the scattering features / retrain the type head with noisy samples.
- **Detection generalises, typing does not.** The model finds 100% of synthetic defects (it learned what a bright compact/elongated region looks like) but can only type them at chance — synthetic texture differs from real TFM speckle, so the texture-based type classifier can't place them.
- **The text probe is a clean pass.** 0% false positives means the model does NOT mistake bright text for a defect — strong evidence it learned defect *texture/shape*, not merely 'bright = defect'. This is the most reassuring result of the three.